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MATERIAL HANDLING MACHINE

Abstract

A working machine having a front and a rear defined by the principal direction of travel, the machine defining a fore-aft centerline, the machine comprising a body mounted on a ground-engaging structure to permit movement of the machine over the ground; an operator cab mounted on the body towards a first side of the centerline; a single lifting arm pivotably mounted towards an aft end of the body, the lifting arm being located within a lifting arm housing, wherein the lifting arm housing comprises a sidewall distal the cab; a drive assembly including a first electric motor for providing power to the ground-engaging structure; a first high voltage electric energy storage unit assembly, including at least one high voltage electric energy storage unit, located aft of the cab; and a second high voltage electric energy storage unit assembly, including at least one high voltage electric energy storage unit, located to a second side of the centerline, opposite the first side of the centerline.

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Background/Summary

FIELD

[0001] This invention relates to a working machine having a lifting arm. More particularly, this invention relates to an electrically powered working machine.

BACKGROUND

[0002] Telehandlers are generally well known and comprise a work machine with a pivoting telescopically extending lifting arm which allows items to be transported between different locations at varying heights with relative ease and flexibility. Telehandlers are often utilized in agriculture, construction or logistics, amongst other sectors.

[0003] Telehandlers are often described in relation to their maximum lifting load and the vertical height a load can be lifted to. Typically, loads and the lifting height will vary depending on the type of machine and industry in which it is used. For example, a compact telehandler may be rated to lift around 1400 kg to a vertical height of 4 m, with a larger agricultural machine lifting around 6000 kg to a vertical height of 8m. However, a large range of rated loads and lifting heights are possible with some machines being configured to lift to 20m or higher and greater weights.

[0004] Another type of material handling machine are known as rotating telehandlers or roto-telehandlers or simply “rotos”. These machines are configured to slew about a main vertical axis such that a main body which carries the working arm is able to rotate with respect to a ground engaging structure and chassis. Such machines are generally larger than conventional telehandlers and can handle larger loads at extended heights, e.g., in excess of 5000 kg to heights above 20 m, although this is not a limitation and compact roto-telehandlers which lift smaller loads to lower heights exist. Roto-telehandlers are typically used when stationary but this is not a limitation.

[0005] Other material handling machines include so-called skid steers which may also be equipped with a telescopic arm. Skid steers generally comprise a compact rigid frame with fixed wheels or tracks which do not pivot relative to the body to effect steering. The respective wheels or tracks on each side of the machine are driven in unison and have independent speed control relative to the other side such that changing the relative speed results in a change in direction. Skid steers generally have limited lifting capacity and height due to the compact nature of the machine.

[0006] A yet further type of material handling machine is a wheeled loading shovel which may also be provided with a telescopically extending shovel arm. Generally, wheeled loading shovels are characterized by an articulating chassis to effect steering. These machines are typically used with a shovel for loading large quantities of materials into a transportation vehicle or around a location and have limited extension.

[0007] Telehandlers of the type of which this disclosure is concerned may thus be differentiated over other types of material handling machine by having steerable wheels which move relative to the chassis, either in two wheel or four-wheel steering modes, and comprise a pivoting working arm or boom which is provided in a fixed relation to the chassis and ground engaging structure. Telehandlers may also be characterized by their lift capacity and height, which may generally be taken to be above 1200 kg and 3 m respectively.

[0008] Working machines are typically diesel-powered. However, there is a drive in the industry to move towards hybrid, electric or hydrogen powered vehicles, particularly where such vehicles are

used indoors. One difficulty with this is removal of the diesel engine, since this significantly alters the weight distribution of the machine and has implications for the stability of the working machine. Counterweight requirements of the machine may be increased in order to provide stability.

[0009] Re-designing the working machine to be powered by an electric motor or motors rather than by a diesel engine can also lead to a requirement for investment in new infrastructure such as assembly lines, and reduced efficiency of assembly and maintenance operations due to the different layout of an electrically-powered machine. Other factors such as cooling must also be taken into consideration. Installation of components can be difficult due to restricted space envelopes.

[0010] The present disclosure seeks to overcome or at least mitigate the problems of the prior art.

SUMMARY

[0011] According to a first aspect of the present disclosure, there is provided a working machine having a front and a rear defined by the principal direction of travel, the machine defining a fore-aft centerline, the machine comprising a body mounted on a ground-engaging structure to permit movement of the machine over the ground; an operator cab mounted on the body towards a first side of the centerline; a single lifting arm pivotably mounted towards an aft end of the body, the lifting arm being located within a lifting arm housing, wherein the lifting arm housing comprises a sidewall distal the cab; a drive assembly comprising a first electric motor for providing power to the ground-engaging structure; a first high voltage electric energy storage unit assembly, comprising at least one high voltage electric energy storage unit, located aft of the cab; and a second high voltage electric energy storage unit assembly, comprising at least one high voltage electric energy storage unit, located to a second side of the centerline, opposite the first side of the centerline.

[0012] Locating the electric energy storage units in such a way addresses the issue of weight distribution in the absence of a diesel engine. The weight of the electric energy storage unit or units in those locations improves stability of the working machine and provides a counterbalance to the lifting arm. The space made available by the absence of a diesel engine is advantageously utilized. Additional power capacity is enabled by providing electric energy storage units in multiple locations. Providing first and second electric energy storage unit assemblies allows flexibility of weight distribution.

[0013] The second electric energy storage unit assembly may be located to the second side of the sidewall of the lifting arm housing.

[0014] The first electric energy storage unit assembly may be mounted to the sidewall.

[0015] Such location of the second electric energy storage unit assembly improves stability of the working machine and provides a counterbalance to the lifting arm. The space made available by the absence of a diesel engine is advantageously utilized.

[0016] The first and second electric energy storage unit assemblies may be remote from one another.

[0017] Separate electric energy storage unit assemblies remote from one another, i.e., separated from one another in detached units rather than adjacent one another, allows flexible mounting arrangements, and so customized weight distribution.

[0018] At least one electric energy storage unit assembly may further comprise a mounting bracket by which said electric energy storage unit assembly is secured to the body.

[0019] Said mounting bracket may comprise a planar portion configured to support an underside of the or each electric energy storage unit.

[0020] The mounting bracket may define a recess or recesses for receiving a fork of a forklift vehicle.

[0021] Such recesses enable easy and secure handling of the electric energy storage unit assembly by a forklift vehicle.

[0022] Such a mounting bracket enables the electric energy storage units to be simply and

efficiently mounted to the machine. Electric energy storage units can be mounted to the mounting bracket prior to assembly of the electric energy storage units on the machine, improving ease of installation of electric energy storage units. The bracket comprising a planar portion is a simple and effective means of supporting the electric energy storage units.

[0023] The second electric energy storage unit assembly mounting bracket may comprise a plurality of mounting positions for receiving an electric energy storage unit. The mounting positions may be linearly arranged with respect to one another, defining a fore-aft longitudinal axis.

[0024] Providing electric energy storage unit mounting positions linearly arranged with respect to one another in a fore-aft direction forms a suitable arrangement for the space envelope provided in a typical machine.

[0025] When the number of mounting positions is greater than the number of electric energy storage units secured by the mounting bracket, each electric energy storage unit may be received within the aft-most mounting position available.

[0026] Positioning the electric energy storage units as far aft as possible is advantageous for weight distribution. In a mounting bracket with capacity for more electric energy storage units than are utilized, it is advantageous to position the electric energy storage units as far aft as the mounting bracket allows.

[0027] The machine may comprise a side enclosure mounted to the side of the body to the second side of the centerline. The second electric energy storage unit assembly may be located within the side enclosure.

[0028] Locating the second electric energy storage unit assembly within a side enclosure makes effective use of available space, and allows access to the second electric energy storage unit assembly, providing ease of access for electric energy storage unit installation and replacement.

[0029] The ground-engaging structure may comprise a rear axle, and the first electric energy storage unit assembly may extend at least in part over the rear axle.

[0030] This location of the first electric energy storage unit assembly optimizes weight distribution.

[0031] There is further provided a working machine having a front and a rear defined by the principal direction of travel, the machine defining a fore-aft centerline, the machine comprising a body mounted on a ground-engaging structure to permit movement of the machine over the ground; an operator cab mounted on the body towards a first side of the centerline; a single lifting arm pivotably mounted towards an aft end of the body, the lifting arm being located within a lifting arm housing, wherein the lifting arm housing comprises a sidewall distal the cab; a drive assembly comprising a first electric motor for providing power to the ground-engaging structure; at least one high voltage electric energy storage unit assembly, comprising at least one high voltage electric energy storage unit, located to a second side of the centerline, opposite the first side of the centerline; and a hydraulic tank, located to the second side of the sidewall of the lifting arm housing.

[0032] Locating the electric energy storage units and the hydraulic tank in such a way addresses the issue of weight distribution in the absence of a diesel engine. The weight of the electric energy storage unit or units in those locations improves stability of the working machine and provides a counterbalance to the lifting arm. The space made available by the absence of a diesel engine is advantageously utilized.

[0033] The ground-engaging structure may comprise a pair of rear wheels, and the hydraulic tank may be shaped so as to fit within a space envelope defined at least in part by one of said wheels.

[0034] The hydraulic tank may comprise a first face at a first angle to a vertical axis of the machine.

[0035] The first angle may be within the range of 19° and 29° to the vertical axis of the machine.

[0036] The first angle may be 24° to the vertical axis of the machine.

[0037] The first face may be proximal said wheel.

[0038] Shaping the hydraulic tank to fit around the space envelope defined by the wheel enables

best use of the available shape, advantageously allowing the most compact possible arrangement of components.

[0039] The hydraulic tank may comprise an enclosure defining a single chamber. The enclosure may comprise first and second wall portions.

[0040] The tank being constructed of first and second wall portions is a simple and effective tank structure.

[0041] The first wall portion may comprise a planar central portion, and opposing arms extending from the central portion and substantially perpendicular to the central portion.

[0042] Such a first wall portion advantageously provides three sides of the tank in a single component, limiting the number of components required and the number of attachments such as welds to be formed during assembly.

[0043] The second wall portion may comprise a planar central portion, and opposing arms extending from the central portion and substantially perpendicular to the central portion. The second wall portion may be complementary to the first wall portion, such that the opposing arms of the second wall portion meet the central portion of the first wall portion.

[0044] The electric energy storage unit assembly may be located to the second side of the sidewall of the lifting arm housing.

[0045] The electric energy storage unit assembly may be mounted to the sidewall.

[0046] Such location of the second electric energy storage unit assembly improves stability of the working machine and provides a counterbalance to the lifting arm. The space made available by the absence of a diesel engine is advantageously utilized.

[0047] The electric energy storage unit assembly may further comprise a mounting bracket by which the or each electric energy storage unit is secured to the body.

[0048] Said mounting bracket may comprise a planar portion configured to support an underside of the or each electric energy storage unit.

[0049] Such a mounting bracket enables the electric energy storage units to be simply and efficiently mounted to the machine. Electric energy storage units can be mounted to the mounting bracket prior to assembly of the electric energy storage units on the machine, improving ease of installation of electric energy storage units. The bracket comprising a planar portion is a simple and effective means of supporting the electric energy storage units.

[0050] The electric energy storage unit assembly mounting bracket may comprise a plurality of mounting positions for receiving an electric energy storage unit. The mounting positions may be linearly arranged with respect to one another, defining a fore-aft longitudinal axis.

[0051] Providing electric energy storage unit mounting positions linearly arranged with respect to one another in a fore-aft direction allows a suitable arrangement for the space envelope provided in a typical machine.

[0052] When the number of mounting positions is greater than the number of electric energy storage units secured by the mounting bracket, each electric energy storage unit may be received within the aft-most mounting position available.

[0053] Positioning the electric energy storage units as far aft as possible is advantageous for weight distribution. In a mounting bracket with capacity for more electric energy storage units than are utilized, it is advantageous to position the electric energy storage units as far aft as the mounting bracket allows.

[0054] The hydraulic tank may be mounted to the body by the electric energy storage unit assembly mounting bracket.

[0055] Such a hydraulic tank commonly has an associated hydraulic pump and a motor for actuating said pump. Mounting the hydraulic tank to the body by a mounting bracket advantageously allows a reduction in vibration, and therefore noise, due to the motor, in comparison to direct mounting of the hydraulic tank and associated motor to the machine body. Advantageously, the number of components used is limited by the multiple applications of the

mounting bracket.

[0056] The machine may comprise a side enclosure mounted to the side of the body to the second side of the centerline.

[0057] The hydraulic tank may be located within the side enclosure.

[0058] Locating the hydraulic tank within a side enclosure makes effective use of available space, and allows easy access to the hydraulic tank for e.g., maintenance.

[0059] The electric energy storage unit assembly may be located within the side enclosure.

[0060] Locating the electric energy storage unit assembly within a side enclosure makes effective use of available space, and allows access to the electric energy storage unit assembly, providing ease of access for electric energy storage unit installation and replacement.

[0061] The machine may further comprise a hydraulic pump for actuation of the lifting arm.

[0062] The hydraulic pump may be mounted to the hydraulic tank.

[0063] The hydraulic pump may be located within the side enclosure.

[0064] A compact arrangement is thus advantageously provided.

[0065] The machine may further comprise a first inverter associated with the first electric motor. The first inverter may be located within the side enclosure.

[0066] The first electric motor may be at least partially located within the side enclosure.

[0067] The machine may further comprise a second electric motor configured for actuation of said hydraulic pump. The second electric motor may be located within the side enclosure.

[0068] The machine may further comprise a second inverter associated with the second electric motor. The second inverter may be located within the side enclosure.

[0069] The machine may further comprise an auxiliary electric energy storage unit for auxiliary systems. The auxiliary electric energy storage unit may be located within the side enclosure.

[0070] The machine may further comprise, located within the side enclosure, a brake fluid reservoir.

[0071] The machine may further comprise, located within the side enclosure, a washer fluid reservoir.

[0072] The machine may further comprise, located within the side enclosure, an isolator.

[0073] The machine may further comprise, located within the side enclosure, a power distribution unit.

[0074] The operator cab may comprise a door on the cab side of the body.

[0075] The machine may comprise an electric energy storage unit charger for charging the or each electric energy storage unit, and a charge port for connection of the charger to an external power source. The charge port may be located to an aft side of the machine. The charge port may be inset from an aft-most point of the machine.

[0076] Location of the charge port at the rear of the machine allows the charge port to be positioned close to an external power source when the machine is parked, without the need to leave space for e.g., opening a side door.

[0077] The machine may comprise a further electric energy storage unit charger for charging the or each electric energy storage unit, and a further charge port for connection of the charger to an external power source.

[0078] Said further charge port may be located to an aft side of the machine.

[0079] Alternatively, said further charge port may be located to the second side of the machine.

[0080] This charge port location allows easy positioning of the machine such that the charge port is close to a power supply.

[0081] The machine may further comprise a valve block mounted to the body, and a valve block mounting arrangement. The valve block mounting arrangement may comprise a resilient mounting.

[0082] Providing a resilient mounting for the valve block advantageously reduces vibration of the valve block during operation of the machine. Reducing vibration leads to a reduction in noise. This is of particular advantage in an electrically-powered machine, where the absence of a diesel engine

increases awareness of other noise.

[0083] The resilient mounting may comprise a resilient washer between the valve block and the body.

[0084] A resilient washer is a simple and effective means of providing a resilient mounting.

[0085] The machine may further comprise a compressor mounted to the body, and a compressor mounting arrangement. The compressor mounting arrangement may comprise a resilient mounting.

[0086] Providing a resilient mounting for the compressor advantageously reduces vibration of the compressor during operation of the machine. Reducing vibration leads to a reduction in noise. This is of particular advantage in an electrically-powered machine, where the absence of a diesel engine increases awareness of other noise.

[0087] The resilient mounting may comprise a resilient washer between the compressor and the body.

[0088] A resilient washer is a simple and effective means of providing a resilient mounting.

[0089] There is further provided a working machine having a front and a rear defined by the principal direction of travel, the machine defining a fore-aft centerline, the machine comprising a body mounted on a ground-engaging structure to permit movement of the machine over the ground; an operator cab mounted on the body towards a first side of the centerline; and a single lifting arm pivotably mounted towards an aft end of the body, the lifting arm being located within a lifting arm housing, wherein the lifting arm housing comprises a sidewall distal the cab.

[0090] The working machine may comprise a first electric motor for providing power to the ground-engaging structure.

[0091] The working machine may comprise at least one electric energy storage unit assembly, comprising at least one high voltage electric energy storage unit.

[0092] The working machine may comprise an electronically applied parking brake.

[0093] The working machine may further comprise an isolator for electrical disconnection of systems of the working machine. The working machine may further comprise an automatic delay system for delaying isolation.

[0094] Advantageously, delay of isolation of the working machine allows the electronically applied parking brake to be reapplied after a predetermined time after the machine is no longer in use, allowing cooling and the resulting shrinkage of brake components to be accounted for.

[0095] There is also provided an electric energy storage unit assembly comprising at least two high voltage electric energy storage units and a mounting bracket by which said electric energy storage unit assembly is secured to a body of a working machine, wherein the mounting bracket defines a recess or recesses for receiving a fork of a forklift vehicle.

[0096] Where the mounting bracket defines at least two recesses for receiving a fork of a forklift vehicle, the recesses may be sized so as to be configured to receive the forks of a forklift vehicle.

[0097] Where the mounting bracket defines at least two recesses for receiving a fork of a forklift vehicle, the recesses may be spaced from one another so as to be configured to receive the forks of a forklift vehicle.

[0098] Such recesses enable easy and secure handling of the electric energy storage unit assembly by a forklift vehicle.

[0099] There is yet further provided a working machine having a front and a rear defined by the principal direction of travel, the machine defining a fore-aft centerline, the machine comprising a body mounted on a ground-engaging structure to permit movement of the machine over the ground; an operator cab mounted on the body towards a first side of the centerline; a single lifting arm pivotably mounted towards an aft end of the body, the lifting arm being located within a lifting arm housing, wherein the lifting arm housing comprises a sidewall distal the cab; a drive assembly comprising a first electric motor for providing power to the ground-engaging structure; a first high voltage electric energy storage unit assembly, comprising at least one high voltage electric energy storage unit, located aft of the cab; and an engine, located to a second side of the centerline,

opposite the first side of the centerline.

[0100] The engine may be a diesel engine.

[0101] Advantageously, a hybrid-powered machine can thus be provided.

[0102] The machine may comprise a side enclosure. The engine may be located within the side enclosure.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0103] Embodiments will now be described by way of example only with reference to the accompanying figures, in which:

[0104] FIG. 1 is a side view of a working machine on horizontal ground;

[0105] FIG. 2 is an isometric view of a body and cab of the machine according to the present disclosure;

[0106] FIG. 3 is an isometric view of a body of the machine according to the present disclosure;

[0107] FIG. 4 is a partial isometric view of an aft portion of the body of the machine according to the present disclosure;

[0108] FIG. 5 is a plan view of the aft portion of the machine shown in FIG. 4;

[0109] FIG. 6 is a partial isometric view of the body of the machine according to the present disclosure;

[0110] FIG. 7 is a partial isometric view of an electric motor of the machine;

[0111] FIG. 8 is a partial isometric view of a side of the body of the machine according to the present disclosure;

[0112] FIG. 9 is a further partial isometric view of the side of the machine shown in FIG. 8;

[0113] FIG. 10 is a further partial isometric view of the machine, with the side enclosure in place;

[0114] FIG. 11 is a partial isometric view of a side of the machine, showing a side enclosure cover in an open position;

[0115] FIG. 12 is a further partial isometric view of the machine;

[0116] FIG. 13 is a side view of a valve block of the machine;

[0117] FIG. 14 is a side view of a compressor of the machine;

[0118] FIG. 15 is a partial side view of a side of the machine;

[0119] FIG. 16 is a partial plan view of a wheel and the body of the machine; and

[0120] FIG. 17 is a partial isometric view of the wheel and the body of the machine of FIG. 16.

DETAILED DESCRIPTION

[0121] In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of various embodiments and the inventive concept. However, those skilled in the art will understand that: the present invention may be practiced without these specific details or with known equivalents of these specific details; that the present invention is not limited to the described embodiments; and, that the present invention may be practiced in a variety of alternative embodiments. It will also be appreciated that well known methods, procedures, components, and systems may not have been described in detail.

[0122] References to vertical and horizontal in the present disclosure should be understood to be in relation to the machine when stood on horizontal ground in a non-working condition. The term axial is generally used in relation to the longitudinal axis of the machine. The term width is generally used in relation to the longitudinal length, that is, transverse to the length.

[0123] With reference to FIG. 1, an embodiment of the teachings includes a machine 1 which may be a load handling machine. In this embodiment the load handling machine is a telescopic handler. In other embodiments the load handling machine may be a skid-steer loader, a compact track loader, a wheel loader, or a telescopic wheel loader, for example. Such machines may be denoted as

off-highway working machines. The machine **1** includes a machine body **2** which may include, for example, an operator's cab **3** from which an operator can operate the machine **1**.

[0124] In an embodiment, the machine **1** has a ground engaging propulsion structure comprising a first axle **A1** and a second axle **A2**, each axle being coupled to a pair of wheels (two wheels **4**, **5** are shown in FIG. **1** with one wheel **4** connected to the first axle **A1** and one wheel **5** connected to the second axle **A2**). The first axle **A1** may be a front axle and the second axle **A2** may be a rear axle. One or both of the axles **A1**, **A2** may be coupled to a motor **20** (see FIG. **5** discussed below) which is configured to drive movement of one or both pairs of wheels **4**, **5**. The wheels may contact a ground surface **H** and rotation of the wheels **4**, **5** may cause movement of the machine with respect to the ground surface. In other embodiments the ground engaging propulsion structure comprises tracks.

[0125] In an embodiment, at least one of the first and second axles **A1**, **A2** is coupled to the machine body **2** by a pivot joint (not shown) located at substantially the center of the axle such that the axle can rock about a longitudinal axis of the machine **1**—thus improving stability of the machine **1** when moving across uneven ground. It will be appreciated that this effect can be achieved in other known manners.

[0126] A load handling apparatus **6**, **7** is coupled to the machine body **2**. The load handling apparatus **6**, **7** may be mounted by a mount **9** to the machine body **2**. In an embodiment, the load handling apparatus **6**, **7** includes a lifting arm **6**, **7**.

[0127] The lifting arm **6**, **7** may be a telescopic arm having a first section **6** connected to the mount **9** and a second section **7** which is telescopically fitted to the first section **6**. In this embodiment, the second section **7** of the lifting arm **6**, **7** is telescopically moveable with respect to the first section **6** such that the lifting arm **6**, **7** can be extended and retracted. Movement of the first section **6** with respect to the second section **7** of the lifting arm **6**, **7** may be achieved by use of an extension actuator **8** which may be a double acting hydraulic linear actuator. In some embodiments, movement of the first section **6** with respect to the second section **7** may be achieved by use of an electric linear actuator, a telescopic extension ram, multiple extension rams, and/or a chain and pulley system. One end of the extension actuator **8** is coupled to the first section **6** of the lifting arm **6**, **7** and another end of the extension actuator **8** is coupled to the second section **7** of the lifting arm **6**, **7** such that extension of the extension actuator **8** causes extension of the lifting arm **6**, **7** and retraction of the extension actuator **8** causes retraction of the lifting arm **6**, **7**. As will be appreciated, the lifting arm **6**, **7** may include a plurality of sections: for example, the lifting arm **6**, **7** may comprise two, three, four or more sections. Each arm section may be telescopically fitted to at least one other section.

[0128] The lifting arm **6**, **7** can be moved with respect to the machine body **2** and the movement is preferably at least in part rotational movement about the mount **9** (about pivot **B** of the lifting arm **6**, **7**). The rotational movement is about a substantially transverse axis of the machine **1**, the pivot **B** being transversely arranged.

[0129] Rotational movement of the lifting arm **6**, **7** with respect to the machine body **2** is, in an embodiment, achieved by use of at least one lifting actuator **10** coupled, at one end, to the first section **6** of the lifting arm **6**, **7** and, at a second end, to the machine body **2**. The lifting actuator **10** is a double acting hydraulic linear actuator, but may alternatively be single acting. In some embodiments, the lifting actuator is an electric linear actuator.

[0130] FIG. **1** shows the lifting arm **6**, **7** positioned at three positions, namely **X**, **Y** and **Z**, with positions **X** and **Y** shown in dashed lines in simplified form. When positioned at position **X** the angle between the lifting arm and a ground level is 55° . This angle is measured with respect to the longitudinal major portion of the lifting arm **6**, **7**, i.e., the part that extends and retracts if the arm is telescopic. In other embodiments, a different measure of the angle may be used, for example an angle defined using a notional line between the pivot **B** and the pivot **D** for the load handling implement (see below). When positioned at position **Y** the angle is 27° . When positioned at

position Z the angle is -5° . 55° and -5° represent the upper and lower limits of angular movement for the machine **1** with stabilizers retracted. The upper limit may be permitted to be increased to, say, 70° when the stabilizers are deployed to contact the ground (see below). Clearly, the lifting arm can be positioned at any angle between these limits. Other machines may have different upper and lower angular limits dependent upon the operational requirements of the machine (maximum and minimum lift height and forward reach etc.) and the geometry of the machine and load handling apparatus (e.g., position of pivot B, dimensions of cranked portion at the distal end of the second section **7** of the lifting arm **6**, **7**). As will be appreciated, when the lifting arm is positioned relatively close to the ground it is at a relatively small angle and when it is positioned relatively remotely from the ground it is at a relatively large or high angle.

[0131] A load handling implement **11** may be located at a distal end of the lifting arm **6**, **7**. The load handling implement **11** may include a fork-type implement which may be rotatable with respect to the lifting arm **6**, **7** about a pivot D, this pivot also being transversely arranged. Other implements may be fitted such as shovels, grabs etc. Movement of the load handling implement **11** may be achieved by use of a double acting linear hydraulic actuator (not shown) coupled to the load handling implement **11** and the distal end of the section **7** of the lifting arm **6**, **7**.

[0132] Off-highway machines **1** of the teachings are configured to transport loads L over uneven ground, i.e., with a load held by the load handling implement **11**, an operator controls the propulsion structure to move the entire machine with the load from one location to another.

[0133] This may be contrasted with machines such as mobile cranes and roto-telehandlers in which a boom is pivotable about both a lateral and an upright axis—i.e., the boom can slew relative to a machine body on a turret or turntable—as well as pivot upwards about the lateral axis. Such machines may be driven to a particular location and are immobilized on four or more stabilizer legs to lift the wheels or other propulsion means entirely off the ground, and to ensure the upright slew axis is absolutely vertically aligned. From that fixed location the machine will move a load from one location to another location using a movements of the boom about the lateral and upright axes. As such, different stability considerations apply to machines in which a boom can also move about an upright axis. Therefore different safety legislation, and consequently different safety systems, are employed on such machines.

[0134] In the illustrated embodiment, the operator cab **3** has a fixed angular orientation with respect to the front and/or rear axles A1 and A2. As shown in FIG. 2, the working machine **1** has a longitudinal axis A3.

[0135] With reference now to FIGS. 3 to 5, the material handling machine **1** of the present disclosure has first and second electric energy storage unit assemblies **40**, **42**. Each electric energy storage unit assembly **40**, **42** includes at least one battery or electric energy storage unit **44**.

[0136] The first electric energy storage unit assembly **40** is located aft of the cab **3**.

[0137] The material handling machine **1** of the present disclosure has a first or cab side S1 of the center line A3, and a second side S2 of the center line A3, opposing the cab side. The second electric energy storage unit assembly **42** is positioned to the second side S2 of the center line A3.

[0138] As referred to herein, the term “side” is used to mean a surface of the working machine that is not the top or bottom with respect to the normal orientation of the machine, and is not the front or back with respect to the direction of travel of the machine over ground.

[0139] The machine **1** having first and second electric energy storage unit assemblies **40**, **42** allows flexibility of positioning of the electric energy storage units **44**, so that weight distribution of the electric energy storage units **44** on the machine **1** can be optimized. The design difficulties created by the removal of a diesel engine can be addressed, allowing a similar body to be used for the machine **1**, and so advantageously limiting re-design of both the machine **1** and assembly lines. Stability of the material handling machine **1** can be optimized, as a counterbalance can be provided to the lifting arm by the electric energy storage unit assemblies **40**, **42**. Advantageously, the counterweight requirements for the machine **1** may be decreased.

[0140] In positioning the second electric energy storage unit assembly **42** to the second side **S2** of the center line **A3**, the space opened up by the removal of a diesel engine is advantageously utilized, as a diesel engine would normally be located to the second side **S2** of the center line.

[0141] In the material handling machine **1** of this disclosure, the second electric energy storage unit assembly **42** is wholly positioned to the second side **S2** of the center line **A3**. That is, the second electric energy storage unit assembly **42** is positioned such that no part of the electric energy storage unit or units **44** included in the second electric energy storage unit assembly **42** is to the first side **S1** of the center line **A3**. In an alternative material handling machine **1**, a majority of the second electric energy storage unit assembly is located to the second side **S2** of the center line **A3**.

[0142] The second electric energy storage unit assembly **42** of the material handling machine **1** of the present disclosure is positioned to the second side **S2** in relation to the sidewall **24_1** of the lifting arm housing **25**. The second electric energy storage unit assembly **42** is in the material handling machine **1** of the present disclosure mounted to the sidewall **24_1**. Mounting of the second electric energy storage unit assembly **42** to the sidewall **24_1** will be described in further detail below.

[0143] In the machine **1** of the present disclosure, the first and second electric energy storage unit assemblies **40**, **42** are remote from one another. That is, the first and second electric energy storage unit assemblies **40**, **42** are located separately to one another on the machine **1**. The first and second electric energy storage unit assemblies **40**, **42** are spaced apart from one another, rather than being in a single array. The first and second electric energy storage unit assemblies **40**, **42** are configured for installation separate to one another.

[0144] Provision of first and second electric energy storage unit assemblies **40**, **42** at different locations of the body **2** enables customization and thus optimization of weight distribution on the material handling machine **1**.

[0145] In addition, provision of the first electric energy storage unit assembly **40** aft of the cab **3** allows the machine to be a hybrid-powered machine. That is, in one embodiment (not shown), the electric energy storage unit assembly **40** is provided in combination with a diesel engine positioned to the second side **S2** of the center line **A3**, e.g., in a side enclosure of the machine. In such an arrangement, the diesel engine would be the prime mover. In this embodiment, the machine includes a hydraulic tank, located aft of the cab.

[0146] As shown most clearly in FIGS. **4** and **8**, each electric energy storage unit assembly **40**, **42** includes a series of electric energy storage units **44** mounted to the machine **1** by a mounting bracket **46**, **51**. The first electric energy storage unit assembly **40** includes in this embodiment two electric energy storage units **44**. In alternative embodiments, the first electric energy storage unit assembly **40** includes one electric energy storage unit **44** or three or more electric energy storage units **44**.

[0147] The first electric energy storage unit assembly **40** includes a mounting bracket **46** having a base plate **46a** upon which the electric energy storage units **44** are supported. The base plate **46a** is secured to a sidewall **50** of the chassis **24**, so as to secure the first electric energy storage unit assembly **40** to the machine **1**.

[0148] In this embodiment, the first electric energy storage unit assembly **40** is further secured to the sidewall **50** by a top plate **46b** which forms part of the mounting bracket **46**. Two side plates **47** serve to secure the electric energy storage units with respect to one another and to the top plate **46b**.

[0149] In this embodiment, the first electric energy storage unit assembly **40** includes at least one pin **48** for improving ease of handling and installation of the first electric energy storage unit assembly **40**. In this embodiment, each pin **48** has a loop **48a** defining an aperture **48b**, by which the pin can be grasped, e.g., manually or by a suitable tool. Each pin **48** of this embodiment is secured to the top plate **46b**, e.g., each pin **48** is received in a corresponding aperture in the top plate **46b** in an interference fit.

[0150] The first electric energy storage unit assembly **40** is assembled as a unit prior to installation,

in order to allow efficient and easy installation of the electric energy storage units **44**.

[0151] In this embodiment, the sidewall **50** is adjacent the lifting arm **6, 7**, such that the first electric energy storage unit assembly **40** is proximal the lifting arm **6, 7**. In alternative embodiments, the first electric energy storage unit assembly **40** is positioned further from the center line **A3**, and from the lifting arm **6, 7**, i.e., further towards the first side **S1**, depending on the required weight distribution.

[0152] The first electric energy storage unit assembly **40** extends towards the rear wheel **5** on the first side **S1**. In this embodiment, the body **2** includes a housing **41** within which the first electric energy storage unit assembly **40** is housed. In this embodiment, as shown in FIGS. **16** and **17**, the housing **41** is shaped to allow the extension of the first electric energy storage unit assembly **40** towards the rear wheel **5**, whilst being shaped to provide a space envelope such that clearance of the wheel **5** is provided as it is turned. The most effective use of the available space can thus be made.

[0153] As shown in FIG. **8**, the mounting bracket **51** of the second electric energy storage unit assembly **42** includes a planar or substantially planar portion **51a** upon which the electric energy storage units **44** are supported. The mounting bracket **51** has side plates **51b** for securing components mounted thereon. The mounting bracket **51** is secured to the sidewall **24_1** so as to mount the second electric energy storage unit assembly **42** to the body of the machine **1**.

[0154] The mounting bracket **51** is shown in further detail in FIG. **15**. In this embodiment, the second electric energy storage unit assembly **42** includes at least one recess **51a** configured to receive a fork of a forklift vehicle, for ease of handling and transportation of the second electric energy storage unit assembly **42**. In this embodiment, the second electric energy storage unit assembly **42** incorporates two such recesses **51a**. In this embodiment, the recesses **51a** are defined by the mounting bracket **51**. The recesses **51a** are sized, and spaced from one another, so as to be configured to receive the forks of a forklift vehicle.

[0155] As shown in FIGS. **8** and **9**, the second electric energy storage unit assembly **42** of this embodiment includes six electric energy storage units **44**. In alternative embodiments, the second electric energy storage unit assembly has five or fewer electric energy storage units **44**, or seven or more electric energy storage units **44**.

[0156] The second electric energy storage unit assembly **42** has a series of mounting positions **52** for electric energy storage units **44**. In the disclosed embodiment, each mounting position **52** has an electric energy storage unit **44** mounted therein. The mounting positions **52** are linearly arranged with respect to one another defining a fore-aft longitudinal axis **X**. It may be that a full complement of electric energy storage units **44**, i.e., as many electric energy storage units as there are mounting positions **52**, is not required for a particular machine. In such a case, the aft-most mounting position **52a** will in one particular embodiment receive the first electric energy storage unit **44**. The second aft-most mounting position **52b** will receive the second electric energy storage unit **44**, and so on. That is, where the number of mounting positions **52** is greater than the number of electric energy storage units **44** in the second electric energy storage unit assembly **42**, each electric energy storage unit **44** is received within the aft-most mounting position **52** available. Advantageously, weight distribution is kept as far aft as is possible, optimizing stability of the machine **1**.

[0157] As with the first electric energy storage unit assembly **40**, the second electric energy storage unit assembly **42** includes at least one pin **48** for improving ease of handling and installation of the first electric energy storage unit assembly **40**. In this embodiment, each pin **48** has a loop **48a** defining an aperture **48b**, by which the pin can be grasped, e.g., manually or by a suitable tool. Each pin **48** of this embodiment is secured to the top plate **46b**.

[0158] The second electric energy storage unit assembly **42** is assembled as a unit prior to installation, in order to allow efficient and easy installation of the electric energy storage units **44**.

[0159] As shown in FIGS. **2** and **3**, the machine **1** of the present disclosure includes a side enclosure **15** in which components are housed. In this embodiment, the second electric energy

storage unit assembly **42** is located within the side enclosure **15**. The side enclosure **15** is mounted to the side of the machine body **2** to the second side **S2** of the center line. Locating the second electric energy storage unit assembly **42** within the side enclosure provides protection to the electric energy storage unit assembly **42** whilst allowing it to be positioned in a suitable location, making effective use of the available space.

[0160] Access to the second electric energy storage unit assembly, e.g., for charging and/or installation and replacement of the electric energy storage units **44**, is easily provided via a cover **15a** of the side enclosure (see FIG. **11**). The cover **15a** is hingedly secured to the sidewall **24_1**, and is configured to be openable in order to provide access to the side enclosure **15**, e.g., for servicing.

[0161] As shown schematically in FIG. **5**, the material handling machine **1** of the present disclosure includes a rear axle **A2** upon which the rear wheels **5** are supported. In this embodiment, the first electric energy storage unit assembly **40** extends in an aft-ward direction over the rear axle **A2**. This location of the first electric energy storage unit assembly improves weight distribution of the material handling machine **1**.

[0162] With reference now to FIGS. **6** and **7**, the motor **20** of the present disclosure is positioned substantially beneath the lifting arm housing **25**. The machine **1** of the present disclosure includes a “dropbox” **21**, i.e., a series of two or more gears with parallel axes configured to transfer power from the motor **20** to the ground engaging structure **A1**, **A2**. In one embodiment, the dropbox is a two-speed dropbox including a two-speed gear arrangement.

[0163] The machine **1** of the present disclosure includes a service brake in the form of a disc brake **23**. In an alternative embodiment, the service brake is a drum brake. In one embodiment (not shown), the machine **1** includes a spring applied, hydraulic release parking brake. In one embodiment (not shown), the machine **1** includes an electronic parking brake, where brake pads are applied by an electric motor. In such an embodiment, the parking brake may be reapplied after a predetermined length of time, in order to allow for shrinkage of components such as a brake disc on cooling.

[0164] In such an embodiment, a time delay to isolation of the machine may be required, in order for reapplication of the parking brake to take place. For example, an instruction is given to a machine operator not to isolate the machine for a predetermined time, or an automatic delay is applied after the instruction to isolate is given.

[0165] The material handling machine **1** of the present disclosure includes a hydraulic tank **60**, best seen in FIGS. **8** and **9**. In this embodiment, the hydraulic tank **60** is positioned to the second side **S2** of the sidewall **24_1** of the lifting arm housing **25**.

[0166] The hydraulic tank **60** is shaped so as to fit within a space envelope defined at least in part by one of the rear wheels **5**. That is, the hydraulic tank **60** has an angled face **62** opposing or proximal to the rear wheel **5** to the second side **S2** of the machine **1**. The face **62** is angled to fit around the wheel **5** such that a lower end **62a** of the face **62** is towards the front of the machine **1** in relation to an upper end **62b** of the face **62**. In this embodiment, the face **62** is at an angle r of substantially 24° to the vertical, i.e., to a vertical axis t of the machine **1** (see FIG. **15**). In alternative embodiments, the face **62** is at an angle r of between 19° and 29° to the vertical. A tank **60** of such a shape makes best use of the available space envelope, allowing the hydraulic tank **60** to fit at the second side of the sidewall **24_1**.

[0167] The hydraulic tank **60** is in this embodiment secured to the machine body **2** at least in part by the mounting bracket **51** of the second electric energy storage unit assembly **42**. A lower, outboard mounting point **60a** of the hydraulic tank **60** is mounted to the planar portion **51a** of the mounting bracket **51**. The tank **60** is resiliently mounted to the bracket **51**, e.g., by means of a resilient washer. The tank **60** is further secured to the machine body **2** by means of an upper mount (not shown) secured to the sidewall **24_1**.

[0168] The hydraulic tank **60** of this embodiment is a single chamber tank. That is, the hydraulic

tank **60** comprises an enclosure that defines a single chamber for holding hydraulic fluid.

[0169] The hydraulic tank **60** of the present disclosure is formed from only two portions, first and second wall portions **64**, **66**. The first and second wall portions **64**, **66** fit together to create the enclosure of the hydraulic tank **60**. Each wall portion **64**, **66** is substantially U shaped in cross-section, such that each wall portion **64**, **66** forms three sides of the tank **60**.

[0170] The first wall portion **64** has a planar central portion **64a** from which opposing arms **64b** extend at an angle of substantially 90°. In this embodiment, the planar portion **64a** forms the angled face **62a**. The opposing arms **64b** provide top and bottom sides of the hydraulic tank **60**.

[0171] The second wall portion **66** has a planar central portion **66a** and opposing arms **66b** extending therefrom substantially 90° to the planar portion **66a**. The arms **66b** form the sides of the hydraulic tank **60**. The central planar portion **66a** forms a front face of the hydraulic tank **60**.

[0172] The opposing arms **64b**, **66b** interlock and adjoin one another, each extending to meet the planar central portion **66a**, **64a** of the other wall portion, to form the enclosure.

[0173] In the present disclosure, the first and second wall portions **64**, **66** are welded to one another to form the hydraulic tank **60**. In this embodiment, the tank **60** is of steel. In alternative embodiments, the tank is of some other suitable material.

[0174] Such an arrangement provides a simple and effective way of forming the hydraulic tank **60**. Advantageously, a limited number of components are required. Manufacture of the hydraulic tank is also relatively simple, as the amount of e.g., welding required to secure the first and second wall portions **64**, **66** to one another is limited.

[0175] In alternative embodiments, the hydraulic tank is formed of some other number of wall portions, and/or of some alternative type of structure.

[0176] In this embodiment, the hydraulic tank is located to the second side **S2** of the side wall **24_1**. The second electric energy storage unit assembly **42** is positioned forward of the hydraulic tank **60**.

[0177] The second electric energy storage unit assembly **42** is mounted to the body **2** via the mounting bracket **51**, as described above. In this embodiment, the hydraulic tank **60** is supported by the mounting bracket **51**.

[0178] The hydraulic tank **60** of this embodiment has a pin or pins **48** such as those described above secured to a top surface thereof, in order to facilitate handling.

[0179] The machine **1** includes a hydraulic pump **68** for actuation of the lifting arm. The machine **1** also includes a second electric motor **70** for actuation of the hydraulic pump **68**. The hydraulic pump **68** is mounted to the hydraulic tank **60**, and in this embodiment is at least partially submerged therein. The second electric motor **70** is supported on the hydraulic tank **60**.

[0180] As the hydraulic tank **60** is secured to the chassis **24** by the mounting bracket **51**, transfer of vibration of the motor to the rest of the machine **1** is advantageously reduced, leading to a reduction in noise levels. That is, vibrations from the accompanying noise from the second electric motor **70** are reduced in comparison to the motor being directly mounted to the chassis.

[0181] Mounting the hydraulic tank **60** via the mounting bracket **51** also advantageously limits the number of components used.

[0182] The hydraulic tank **60** of the present disclosure is located within the side enclosure **15**. Advantageously, such location of the hydraulic tank **60** allows effective use of the available space, e.g., the space opened up by the removal of a diesel engine of the machine design.

[0183] Also housed within the side enclosure **15**, as shown in FIG. **3**, is an auxiliary 12V electric energy storage unit **74**, for providing power to auxiliary systems such as lights. The 12V electric energy storage unit **74** is supported on a bracket **76** that is secured to the sidewall **24_1**.

[0184] The machine **1** includes a power distribution unit **75** and an isolator **77**, both housed within the side enclosure **15**. The power distribution unit **75** is supported by the bracket **76**. The isolator **77** is mounted to the sidewall **24_1**. Location of the isolator **77** in the side enclosure **15** improves use of access and maintenance.

[0185] The machine **1** includes a first inverter **78** and a second inverter **79** (see FIG. **9**) in a subassembly. The first inverter **78** is associated with the first electric motor **20**. The second inverter **79** is associated with the second electric motor **70**. Each inverter **78**, **79** has an associated heat sink. The inverters **78**, **79** are supported on the bracket **76**. As shown in FIG. **10**, the inverter subassembly includes an inverter cover **80** that covers the inverters **78**, **79**.

[0186] As shown in FIG. **10**, also mounted within the side enclosure **15** are service points for brake fluid **81**, hydraulic fluid **83** and washer fluid **93**. Advantageously, mounting such service points in the side enclosure **15** provides easy access for maintenance.

[0187] FIG. **11** shows the side enclosure **15** with the cover **15a** in an open position. In this embodiment, the cover is configured to provide integrated cable storage. That is, an inside face **15b** of the cover **15a** is shaped to provide slots for receiving charger cables and the like. Advantageously, such storage provides easy access, increasing efficiency of e.g., vehicle charging.

[0188] With reference now to FIGS. **12** to **14**, the machine **1** further includes a valve block **84**. The valve block **84** includes electro-hydraulic valves **85** for controlling hydraulic flow. In the machine **1** of the present disclosure, the valve block is positioned between the cab **3** and the first electric energy storage unit assembly **40**. As shown in FIG. **12**, the valve block **84** is mounted to the sidewall **50**.

[0189] FIG. **13** shows a mounting arrangement **86** by which the valve block **84** is secured to the body **2** in further detail. The mounting arrangement **86** includes multiple mounting points **87**. Each mounting point **87** includes a bolt **87a** that extends through an aperture defined by a projection **87b** of the valve block **84**. The bolts **87a** are threaded, and are received in apertures in the sidewall **50**, and each is secured by a nut (not shown) to secure the valve block **84** to the sidewall **50**.

[0190] The mounting arrangement **86** includes a resilient mounting **87c**. Advantageously, the resilient mounting **87c** reduces vibration of the valve block **84** during machine operation, and so reduces noise. Each mounting point **87** includes a resilient washer **87c** between the projection **87b** and the sidewall **50**. In this embodiment, the resilient washers **87c** are of silicone rubber.

[0191] In alternative embodiments, the washers are of some other suitable resilient material.

[0192] In alternative embodiments, the valve block **84** is mounted to the body by some other suitable means.

[0193] The machine **1** further includes an air conditioning compressor **88** as shown in FIGS. **12** and **14**. The compressor **88** is electrically driven. As shown in FIG. **14**, the compressor **88** includes a mounting arrangement **89**. The mounting arrangement **89** includes a bracket **90** to which the compressor **88** is secured. In this embodiment, the compressor **88** is secured to the bracket by bolts **91**. The mounting arrangement **89** includes multiple mounting points **92** at which the bracket **90** is secured to the machine body **2**. Each mounting point **92** includes a bolt **92a** that extends through an aperture in the bracket **90**. The bolts **92a** are threaded, and are secured by a nut (not shown) in the side wall **50**.

[0194] The mounting arrangement **92** includes a resilient mounting **92b**. Advantageously, the resilient mounting **92b** reduces vibration of the compressor **88** during operation, and so reduces noise. Each mounting point **92** includes a resilient washer **92b** between the bracket **90** and the side wall **50**. In this embodiment, the resilient washers **92b** are of silicone rubber. In alternative embodiments, the washers are of some other suitable resilient material.

[0195] In alternative embodiments, the compressor **88** is mounted to the body by some other suitable means.

[0196] The machine **1** of the present disclosure includes an electric energy storage unit charger and charge port **82** (see FIG. **4**). The charge port **82** is externally accessible by a door. The charger and charge port **82** are located to the aft of the machine **1**, i.e., such that the charge port **82** is accessible from the aft end of the machine **1**. The charge port **82** is inset to protect the charge port **82** from external damage. The machine **1** may include a second charger and charge port, e.g., a charger of different current, facilitating charging of the electric energy storage units at different speeds.

[0197] Advantageously, much of the machine body 2 of this embodiment is similar to that of a known diesel power machine, keeping redesign and assembly costs to a minimum.

[0198] The one or more embodiments are described above by way of example only and it will be appreciated that variations are possible without departing from the scope of protection afforded by the pending claims.

Claims

1. A working machine having a front and a rear defined by the principal direction of travel, the machine defining a fore-aft centerline, the machine comprising: a body mounted on a ground-engaging structure to permit movement of the machine over the ground; an operator cab mounted on the body towards a first side of the centerline; a single lifting arm pivotably mounted towards an aft end of the body, the lifting arm being located within a lifting arm housing, wherein the lifting arm housing comprises a sidewall distal the cab; a drive assembly comprising a first electric motor for providing power to the ground-engaging structure; a first high voltage electric energy storage unit assembly, comprising at least one high voltage electric energy storage unit, located aft of the cab; and a second high voltage electric energy storage unit assembly, comprising at least one high voltage electric energy storage unit, located to a second side of the centerline, opposite the first side of the centerline.
2. The machine of claim 1, wherein the second electric energy storage unit assembly is located to the second side of the sidewall of the lifting arm housing; optionally wherein the first electric energy storage unit assembly is mounted to the sidewall; optionally wherein the first and second electric energy storage unit assemblies are remote from one another.
3. (canceled)
4. The machine of claim 1, wherein at least one electric energy storage unit assembly further comprises a mounting bracket by which said electric energy storage unit assembly is secured to the body; optionally wherein said mounting bracket comprises a planar portion configured to support an underside of the or each electric energy storage unit, optionally wherein the mounting bracket defines a recess or recesses for receiving a fork of a forklift vehicle.
5. The machine of claim 4, wherein the second electric energy storage unit assembly mounting bracket comprises a plurality of mounting positions for receiving an electric energy storage unit, wherein the mounting positions are linearly arranged with respect to one another, defining a fore-aft longitudinal axis.
6. The machine of claim 5, wherein, when the number of mounting positions is greater than the number of electric energy storage units secured by the mounting bracket, each electric energy storage unit is received within the aft-most mounting position available.
7. (canceled)
8. The machine of claim 1, wherein the ground-engaging structure comprises a rear axle, and wherein the first electric energy storage unit assembly extends at least in part over the rear axle.
9. A working machine having a front and a rear defined by the principal direction of travel, the machine defining a fore-aft centerline, the machine comprising: a body mounted on a ground-engaging structure to permit movement of the machine over the ground; an operator cab mounted on the body towards a first side of the centerline; a single lifting arm pivotably mounted towards an aft end of the body, the lifting arm being located within a lifting arm housing, wherein the lifting arm housing comprises a sidewall distal the cab; a drive assembly comprising a first electric motor for providing power to the ground-engaging structure; at least one high voltage electric energy storage unit assembly, comprising at least one high voltage electric energy storage unit, located to a second side of the centerline, opposite the first side of the centerline; and a hydraulic tank, located to the second side of the sidewall of the lifting arm housing.
10. The machine of claim 9, wherein the ground-engaging structure comprises a pair of rear

wheels, and wherein the hydraulic tank is shaped so as to fit within a space envelope defined at least in part by one of said wheels, and/or wherein the hydraulic tank comprises a first face at a first angle to a vertical axis of the machine, wherein the first angle is within the range of 19° and 29° to the vertical axis of the machine, optionally wherein the first angle is 24° to the vertical axis of the machine, optionally wherein the first face is proximal said wheel.

11. The machine of claim 9, wherein the hydraulic tank comprises an enclosure defining a single chamber; wherein the enclosure comprises first and second wall portions.

12. The machine of claim 11, wherein the first wall portion comprises a planar central portion, and opposing arms extending from the central portion and substantially perpendicular to the central portion; optionally wherein the second wall portion comprises a planar central portion, and opposing arms extending from the central portion and substantially perpendicular to the central portion, wherein the second wall portion is complementary to the first wall portion, such that the opposing arms of the second wall portion meet the central portion of the first wall portion.

13. (canceled)

14. The machine of claim 9, wherein the electric energy storage unit assembly is located to the second side of the sidewall of the lifting arm housing; optionally wherein the electric energy storage unit assembly is mounted to the sidewall.

15. The machine of claim 9, wherein the electric energy storage unit assembly further comprises a mounting bracket by which the or each electric energy storage unit is secured to the body; optionally wherein said mounting bracket comprises a planar portion configured to support an underside of the or each electric energy storage unit; optionally wherein the hydraulic tank is mounted to the body by the electric energy storage unit assembly mounting bracket.

16. The machine of claim 15, wherein the electric energy storage unit assembly mounting bracket comprises a plurality of mounting positions for receiving an electric energy storage unit, wherein the mounting positions are linearly arranged with respect to one another, defining a fore-aft longitudinal axis; optionally wherein when the number of mounting positions is greater than the number of electric energy storage units secured by the mounting bracket, each electric energy storage unit is received within the aft-most mounting position available.

17. (canceled)

18. (canceled)

19. The machine according to claim 9, comprising a side enclosure mounted to the side of the body to the second side of the centerline, wherein the hydraulic tank is located within the side enclosure; optionally further comprising an auxiliary electric energy storage unit for auxiliary systems, wherein the auxiliary electric energy storage unit is located within the side enclosure; optionally wherein the machine further comprises, located within the side enclosure, a brake fluid reservoir, and/or a washer fluid reservoir, and/or an isolator, and/or a power distribution unit.

20. The machine according to claim 9, comprising a side enclosure mounted to the side of the body to the second side of the centerline, wherein the electric energy storage unit assembly is located within the side enclosure; optionally further comprising a hydraulic pump for actuation of the lifting arm, wherein the hydraulic pump is mounted to the hydraulic tank; optionally wherein the hydraulic pump is located within the side enclosure.

21. (canceled)

22. The machine of claim 19, wherein the machine further comprises a first inverter associated with the first electric motor, wherein the first inverter is located within the side enclosure; and/or wherein the first electric motor is at least partially located within the side enclosure.

23. The machine of claim 22, further comprising: a hydraulic pump for actuation of the lifting arm, wherein the hydraulic pump is mounted to the hydraulic tank; and a second electric motor configured for actuation of said hydraulic pump, wherein the second electric motor is located within the side enclosure; optionally wherein the machine further comprises a second inverter associated with the second electric motor, and wherein the second inverter is located within the side

enclosure.

24. (canceled)

25. (canceled)

26. The machine of claim 1, wherein the machine comprises an electric energy storage unit charger for charging the or each electric energy storage unit, and a charge port for connection of the charger to an external power source, and wherein the charge port is located to an aft side of the machine; and/or wherein the charge port is inset from an aft-most point of the machine; optionally wherein the machine comprises a further electric energy storage unit charger for charging the or each electric energy storage unit, and a further charge port for connection of the charger to an external power source; optionally wherein said further charge port is located to an aft side of the machine, or wherein said further charge port is located to the second side of the machine.

27. (canceled)

28. The machine of claim 1, further comprising a valve block mounted to the body, and a valve block mounting arrangement, wherein the valve block mounting arrangement comprises a resilient mounting, optionally wherein the resilient mounting comprises a resilient washer between the valve block and the body.

29. The machine of claim 1, further comprising a compressor mounted to the body, and a compressor mounting arrangement, wherein the compressor mounting arrangement comprises a resilient mounting, optionally wherein the resilient mounting comprises a resilient washer between the compressor and the body.
